



Subnetting quick reference

LearnHub course reference: Networking Fundamentals

Common IPv4 prefixes

Prefix	Subnet mask	Addresses	Usable hosts
/8	255.0.0.0	16,777,216	16,777,214
/16	255.255.0.0	65,536	65,534
/20	255.255.240.0	4,096	4,094
/22	255.255.252.0	1,024	1,022
/23	255.255.254.0	512	510
/24	255.255.255.0	256	254
/25	255.255.255.128	128	126
/26	255.255.255.192	64	62
/27	255.255.255.224	32	30
/28	255.255.255.240	16	14
/29	255.255.255.248	8	6
/30	255.255.255.252	4	2

Formulas

- Host bits = 32 – prefix length
- Addresses in the subnet = $2^{\text{host bits}}$
- Usable hosts = addresses – 2 (network and broadcast)
- Block size in the interesting octet = 256 – mask value

Worked example: 192.168.10.77/26

Mask	255.255.255.192 (block size 256 – 192 = 64)
Subnets	.0, .64, .128, .192
Network address	192.168.10.64
First usable host	192.168.10.65
Last usable host	192.168.10.126
Broadcast address	192.168.10.127

Private address ranges (RFC 1918)

Range	Prefix
10.0.0.0 – 10.255.255.255	10.0.0.0/8
172.16.0.0 – 172.31.255.255	172.16.0.0/12
192.168.0.0 – 192.168.255.255	192.168.0.0/16

Special addresses

127.0.0.1	Loopback (this computer)
169.254.0.0/16	Link-local: no DHCP server answered
0.0.0.0/0	Default route

Troubleshooting commands

ipconfig / ip addr	Show your address and mask
ping	Test reachability
tracert / traceroute	Show each router on the path
nslookup	Test DNS name resolution

LearnHub course reference. IPv4 only; IPv6 uses 128-bit addresses and different conventions.